

# Tutorial : how to create a watchmap filter with Lasair

Rubin data is distributed by online brokers, under the format of *enriched alerts* : some ascii files with metadata on the detected transient, and a couple crops of the frames that raised the alert.

These brokers are open-access, and we are going to use them to monitor the localisation region of a GW event.

In this document we shall use Lasair, an UK-based broker with ergonomic watchmap-making tools.

## 1) Create your account

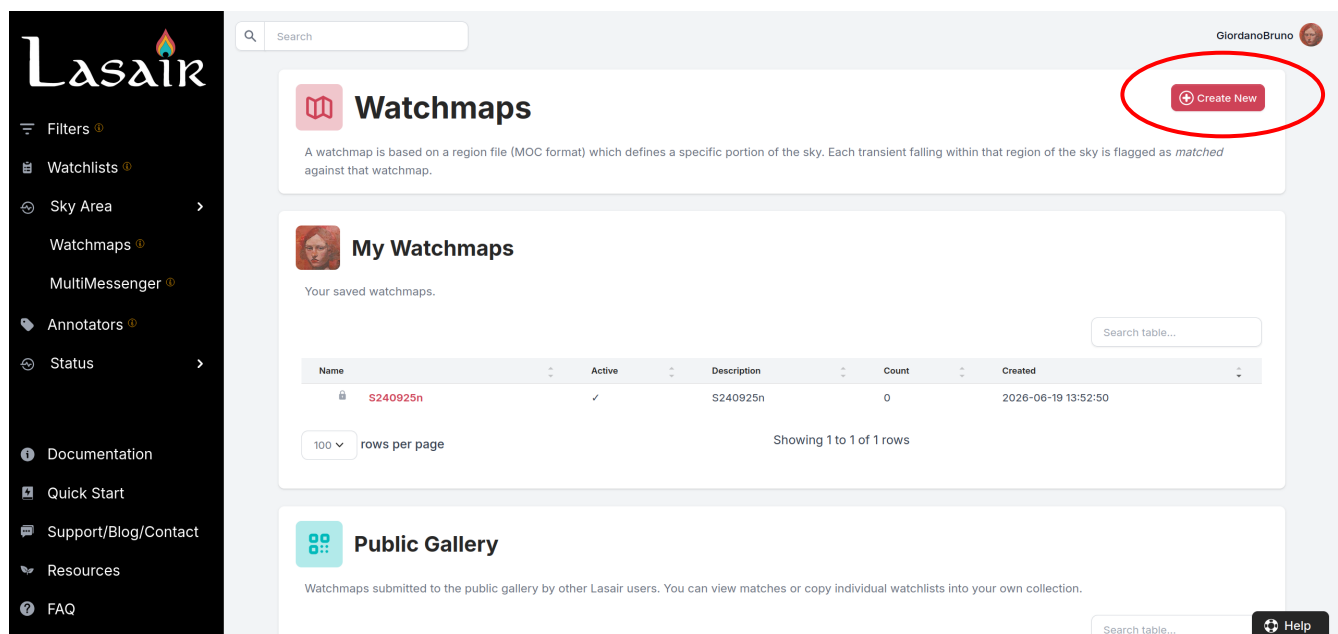
In order to make filters and access Rubin data, you have to create your Lasair account. Register on <https://lasair.lsst.ac.uk/register/> .

## 2) [Optional] : Create a first filter

Once logged in, you are all set for fetching LSST alerts. If you like, try creating a first filter following the instructions on the Lasair website here <https://lasair-lsst.readthedocs.io/en/main/quickstart.html> and get comfortable with the broker and alerts format.

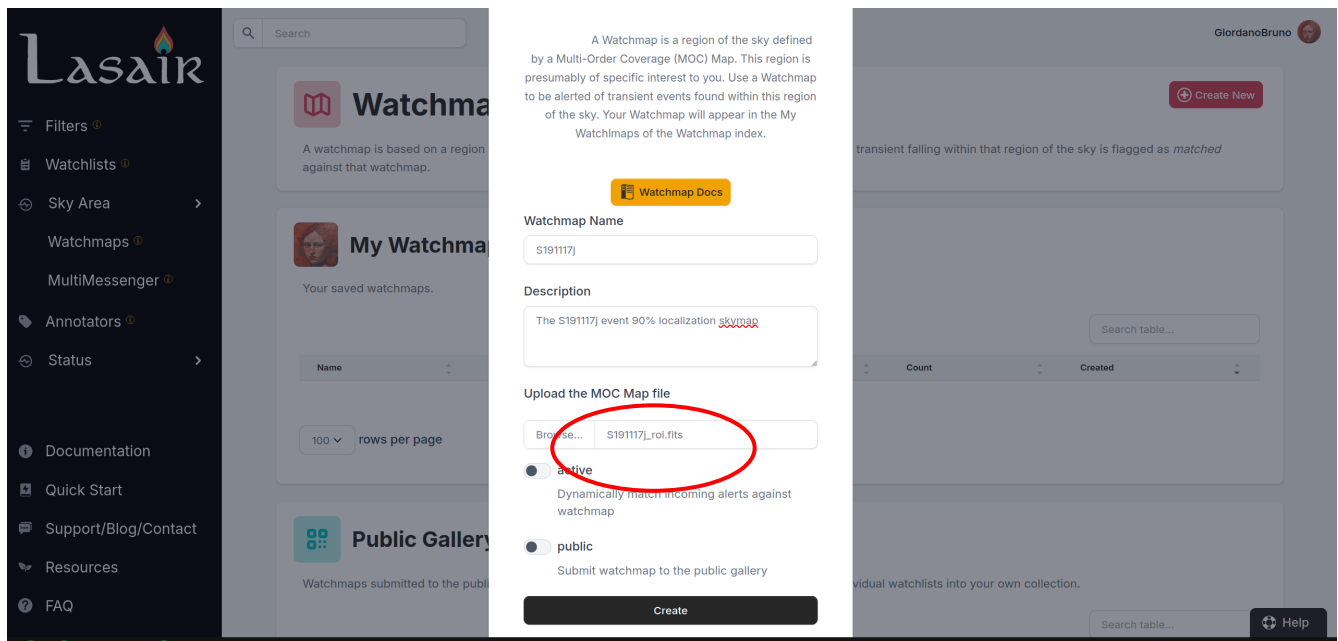
## 3) Create a watchmap

Watchmaps are regions of interest in the sky from which you receive alerts via the broker. In Lasair, you can configure a watchmap prior integrating it into a filter. First, create the watchmap fits file using e.g. *eventVisualisation.py*. Then, load the file to the Lasair wizard as indicated on the screenshots below.



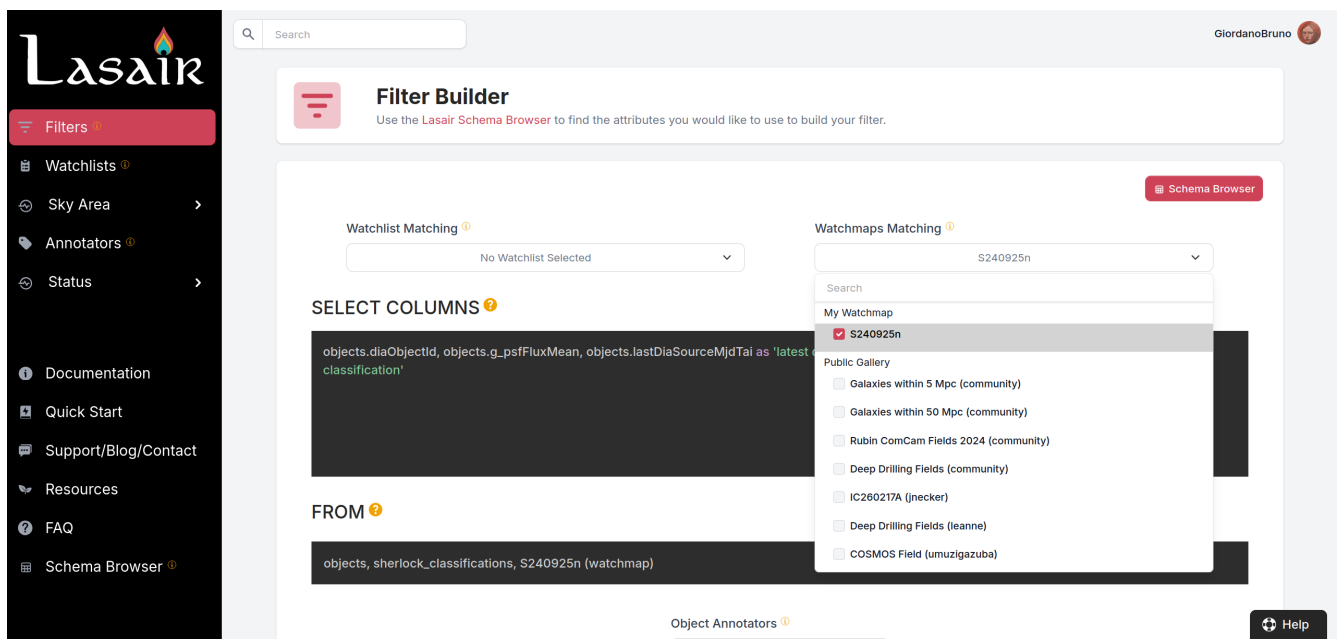
The screenshot shows the Lasair web interface. On the left is a dark sidebar with navigation links: Filters, Watchlists, Sky Area, Watchmaps, MultiMessenger, Annotators, Status, Documentation, Quick Start, Support/Blog/Contact, Resources, and FAQ. The main content area is light gray. At the top right, the user's name 'GiordanoBruno' is displayed next to a profile icon. Below this, the 'Watchmaps' section is highlighted with a red circle around the 'Create New' button. The 'Watchmaps' section includes a description: 'A watchmap is based on a region file (MOC format) which defines a specific portion of the sky. Each transient falling within that region of the sky is flagged as *matched* against that watchmap.' Below this is the 'My Watchmaps' section, which shows a table of saved watchmaps. The table has columns: Name, Active, Description, Count, and Created. One watchmap is listed with the name 'S240925n', which is highlighted in red, and is active. The table also includes a search bar and a 'rows per page' selector. At the bottom of the main content area is the 'Public Gallery' section, which includes a description and a search bar. A 'Help' button is located at the bottom right of the interface.

| Name     | Active | Description | Count | Created             |
|----------|--------|-------------|-------|---------------------|
| S240925n | ✓      | S240925n    | 0     | 2026-06-19 13:52:50 |



## 4) Create a filter associated to the watchmap

Now your watchmap is ready, integrate it to a filter by using the filter wizard.



You can choose to be informed by email of the alerts sent by LSST that match our filter. Otherwise, you can always visualize them in the browser by clicking 'Run Filter'.

Clear skies !